

SHALINI H

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EDUCATION

B.E. in Information Science and Engineering – *The National Institute of Engineering, Mysore (2024-2028)* | CGPA: 8.75/10

Relevant Coursework: Data Structure and Algorithms, Advanced Java, Design and Analysis of Algorithm, Graph Theory, Operating System, Database Management System

TECHNICAL SKILLS

Programming Languages: Java, Python, C, TypeScript, JavaScript

Frontend and Web: React.js, HTML5, CSS3, Tailwind CSS, Web Audio API

Databases and Cloud: MySQL

Tools: Git/Github, Vite, Vitest

Computer Science Fundamentals: Object-Oriented Programming, DSA, Graph Algorithms

PROJECTS

CookPilot – Local-First Cooking Decision Engine (*October 2026*) | *React, TypeScript, Vite, Vitest, Web Audio API*

- Built a zero-backend engine that ranks 20+ recipes from pantry items via weighted scoring, with a state-machine cooking cockpit, timers, and portion scaling.
- Engineered decoupled TypeScript engines for rule-based dish diagnosis and a state-aware rescue mode with what-if simulation, validated with 79 Vitest tests.

AlgoTrace – Interactive Algorithm Execution Visualizer (*October 2026*) | *React, TypeScript, Vite, Tailwind CSS, Vitest*

- Built a visualizer for 17 sorting, searching, and graph algorithms with step playback, scrubbable timeline, synced pseudocode, and live BFS/DFS/Dijkstra panels.
- Architected a pure-function engine that precomputes immutable step traces for lossless backward navigation, backed by input validation and 117 Vitest tests.

CortX – Real-Time Hospital Triage & Queue Management System (*May 2026*) | *HTML5, CSS3, JavaScript*

- Built a zero-backend hospital platform automating triage, doctor assignment, and appointments/inventory across 5 modules with a live waiting-room display.
- Engineered a priority queue with auto specialist matching and rolling-average ETAs from the last 25 logs, synced across tabs via BroadcastChannel and Storage events.

ACHIEVEMENTS

- **Selected for Smart India Hackathon 2026 (College Level) - BharatResQ:** Built the full-stack prototype for AI-based landslide risk monitoring in the Northeast Region of India.